

Day 1: Introduction to Networking and the OSI Model

Assignment Answers

Question 1: Wireshark capture - protocols seen

Steps: Install Wireshark, select the main interface (Wi-Fi / Ethernet), click *Start Capturing*, open instagram.com in a browser, then click *Stop*. Typical protocols that appear in the Protocol column are:

Protocol	What it does
DNS	Converts the name instagram.com into an IP address
TCP	Reliable connection setup (SYN, SYN-ACK, ACK) and data delivery
TLS / TLSv1.2 / TLSv1.3	Encrypts the data (Client Hello, Server Hello, certificate exchange)
QUIC / UDP	Fast transport used by modern HTTPS (HTTP/3)
HTTP / HTTP2	Web requests and responses (mostly encrypted inside TLS)
ARP	Finds the MAC address of the router on the local network
ICMP / ICMPv6, MDNS, SSDP	Network diagnostics and local device discovery

Note: the exact list depends on your own capture; the above is what is commonly observed.

Question 2: Mapping 3 packets to OSI layers

Packet (Protocol / Info)	OSI Layer	Reason
DNS - Standard query A instagram.com	Layer 7 - Application	DNS is an application-level service used by the browser to resolve names.
TCP - [SYN] 52344 → 443	Layer 4 - Transport	TCP uses port numbers and handles connection setup and reliable delivery.
TLSv1.3 - Client Hello	Layer 6 - Presentation	TLS handles encryption of data before it is sent.
ARP - Who has 192.168.1.1?	Layer 2 - Data Link	ARP works with MAC addresses on the local network.

Other examples: HTTP = Layer 7, IP (IPv4/IPv6) / ICMP = Layer 3 Network, Ethernet II frame = Layer 2.

Question 3: OSI Model diagram with real-world examples

Layer	Name	Real-world example from daily apps
7	Application	WhatsApp, Chrome, Instagram app, Gmail
6	Presentation	Instagram photo compression (JPEG), HTTPS/TLS encryption
5	Session	Staying logged in to Instagram / WhatsApp Web session
4	Transport	TCP for chat messages, UDP for WhatsApp voice/video calls
3	Network	IP addressing and routing your data across the internet
2	Data Link	Wi-Fi (802.11) and MAC addresses on your phone/laptop
1	Physical	Wi-Fi radio signals, mobile data, LAN cables, fiber

Memory trick (top to bottom): All People Seem To Need Data Processing. Data moves **down** the layers on the sender's device (encapsulation) and **up** the layers on the receiver's device (decapsulation).

Question 4: How a WhatsApp message travels through the OSI layers

Suppose I type "Hi" and press Send. Here is the journey:

Layer 7 - Application: I type the message in the WhatsApp app and tap Send. WhatsApp creates the message with details like the receiver's number and timestamp. This is where the user interacts with the network service.

Layer 6 - Presentation: The message is converted into a format suitable for transmission and encrypted (WhatsApp uses end-to-end encryption). Media such as photos is compressed. So only the receiver can read it.

Layer 5 - Session: A session with WhatsApp's servers is opened and kept alive, so the app stays connected and the message is part of an ongoing conversation.

Layer 4 - Transport: The data is broken into segments. TCP adds source and destination port numbers (e.g. 443), numbers the segments, and makes sure lost segments are re-sent.

Layer 3 - Network: Each segment is placed in a packet with the sender's IP and the WhatsApp server's IP. Routers use these IP addresses to find the best path across the internet.

Layer 2 - Data Link: Packets are wrapped into frames with MAC addresses so they move from my phone to the Wi-Fi router, and then hop to hop.

Layer 1 - Physical: The frames become radio waves (Wi-Fi/4G/5G), electrical signals or light pulses in fiber cables - actual bits (0s and 1s) travelling.

At the receiver's side: the process is reversed. The phone receives the signal (Layer 1), reads the frame (2), checks the IP packet (3), reassembles the segments in order (4), maintains the session (5), decrypts the data (6), and finally WhatsApp shows "Hi" on the screen (7).